TB Vocoder

DETAILED ENGLISH USER MANUAL

A multi-mode channel vocoder with 8-64 bands, internal multi-voice carrier, optional sidechain carrier, spectral imprint controls, and five transformation types.

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Host-visible endpoints documented: 25

1. Product purpose

A multi-mode channel vocoder with 8-64 bands, internal multi-voice carrier, optional sidechain carrier, spectral imprint controls, and five transformation types.

Classic vocoders can require a separate synth track and provide little control over intelligibility. TB Vocoder can use its internal carrier for immediate operation or accept a sidechain, while analysis range, envelope timing, imprint, whitening, sibilance, and carrier design remain independently adjustable.

What result it creates

- Classic robotic speech and talk-box tone
- Whisper, Morph, Ring Mod, and Invert transformations
- Internal carrier from 2 to 100 voices
- Detailed control over articulation, band resolution, width, and sibilance

Signal flow

Modulator input -> analysis filter bank and envelopes -> internal or sidechain carrier -> selected Vocoder/Whisper/Morph/Ring Mod/Invert process -> spectral imprint and output shaping -> Mix

2. Complete feature guide

Type

Vocoder applies channel envelopes, Whisper replaces pitch with noise-like energy, Morph blends spectral identities, Ring Mod multiplies structures for metallic sidebands, and Invert creates an inverse spectral relationship.

Bands and analysis range

Bands sets resolution from 8 to 64. Range Low and Range High restrict the analyzed spectrum. Fewer bands sound classic and coarse; more bands improve intelligibility at higher CPU cost.

Envelope timing

Attack follows new articulation; Release holds or releases each band. Fast timing is crisp, while slower timing is smoother and more legato.

Internal carrier

Carrier Tune, Resonance, Count, Color, Tone, and Width build the synth source. Count ranges from a small focused stack to a large cloud.

Sidechain

Enable Sidechain to use the external carrier input when the host provides it. Routing must be configured in the DAW.

Imprint and clarity

Imprint sets modulator transfer strength. Detail, Pressure, Whiten, Floor, Sibilance, Warp, and Shift shape clarity, noise floor, formant movement, and articulation.

Mix and Output

Mix blends the transformed result with the original modulator. Output compensates the final level. Programs cover broadcast clarity, robots, choirs, whispers, and broken machines.

3. Quick start

1. Start with Type Vocoder, Sidechain off, Mix 100 percent.

2. Set Bands to 32 and choose the useful Range Low/High.

3. Tune Attack and Release for intelligibility.

4. Adjust Carrier Tune, Count, Color, Resonance, and Width.

5. Use Imprint, Detail, Whiten, and Sibilance to clarify speech.

6. Try another Type only after the basic carrier works.

7. Set Output and final Mix.

4. Practical workflows

Clear robot voice

1. Use Vocoder with 32-64 bands.

2. Use fast Attack and medium Release.

3. Set Imprint high and add Sibilance.

4. Use a bright internal carrier with moderate resonance.

Expected result: Speech remains understandable while becoming strongly synthetic.

External musical carrier

1. Route a synth to the plug-in sidechain.

2. Enable Sidechain.

3. Match analysis range to the synth and voice.

4. Use Mix at 100 percent on the effect return.

Expected result: The voice articulates the rhythm and harmony of the external carrier.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- Sidechain availability and bus naming vary by DAW.
- More bands are not always more musical.
- High carrier counts, width, and resonance can create large peaks.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 25 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Attack VST3 ID 740224584	0.1 to 200.0	3.0	Automatable	Sets how quickly the associated detector, envelope, grain, or dynamics stage responds at onset.
Bands VST3 ID 93503710	8 to 64	32	Automatable	Host-automatable control for Bands. Its full range and default are shown in this table; use it with the related feature section above.
Bypass VST3 ID 1652125811	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Carrier Count VST3 ID 1948654839	2 to 100	3	Automatable	Host-automatable control for Carrier Count. Its full range and default are shown in this table; use it with the related feature section above.
Carrier Resonance VST3 ID 1112088630	0.0 to 100.0	50.0	Automatable	Host-automatable control for Carrier Resonance. Its full range and default are shown in this table; use it with the related feature section above.
Carrier Tune VST3 ID 1379506170	-24 to 24	0	Automatable	Sets the principal frequency, note, or spectral center used by this function.
Color VST3 ID 1948646219	0.0 to 100.0	30.0	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Detail VST3 ID 812259409	0.0 to 100.0	65.0	Automatable	Sets how sensitively the associated analysis or transfer stage responds to source detail.
Floor VST3 ID 97526796	-90 to -30	-70	Automatable	Sets how sensitively the associated analysis or transfer stage responds to source detail.
Imprint VST3 ID 1926118409	0.0 to 100.0	100.0	Automatable	Sets how strongly the named process affects the signal.
Mix VST3 ID 108124	0.0 to 100.0	100.0	Automatable	Sets the balance between the original signal and the complete processed path.
Output VST3 ID 1141971201	-18.0 to 6.0	0.0	Automatable	Sets or limits the final output level after the plug-in's main processing.
Pressure VST3 ID 871241285	0.0 to 100.0	25.0	Automatable	Sets how sensitively the associated analysis or transfer stage responds to source detail.

Parameter	Range / options	Default	Host status	Function
Program VST3 ID 1886548852	Init, Classic Talk Box, Robot Choir, Formant Glass, Low Iron Chant, Ringed Metal, Air Whisper, Hollow Invert, 100 Voice Cloud, Clean Broadcast, Chrome Robot, Pocket Robot, Giant Robot, Rust Robot, Neon Robot, Servo Choir, Data Droid, Broken Android, Wide Machine, Tiny Circuit, Breath Print, Ghost Air, Frost Whisper, Sand Whisper, Secret Radio, Silver Breath, Dark Hiss, Wide Whisper, Close Whisper, Sibilant Cloud, Crystal Formant, Velvet Morph, Glass Choir, Alien Vowel, Rubber Formant, Liquid Mouth, Formant Lift, Formant Drop, Wide Morph, Micro Formants, Chrome Ring, Tin Pulse, Steel Teeth, Bell Circuit, Cold Motor, Broken Metal, Razor Ring, Low Gear, Wide Alloy, Nano Ring, Hollow Core, Reverse Vowel, Empty Shell, Dark Negative, Bright Invert, Thin Void, Wide Hollow, Invert Bass, Ghost Cavity, Crushed Invert, Sub Chant, Monastery Iron, Deep Machine, Low Formant, Bass Drone Print, Octave Below, Dark Stack, Slow Temple, Heavy Vowel, Underworld Choir, Tight Gate, Percussive Print, Fast Chopper, Snare Robot, Drum Carrier, Pluck Tracker, Staccato Glass, Slow Bloom, Pumped Motion, Rhythm Invert, Stereo Cloud, 100 Voice Aurora, Wide Choir, Detuned Sky, Soft Halo, Bright Horizon, Dark Atmosphere, Cloud Shift Up, Cloud Shift Down, Infinite Wash, OTT Talk, OTT Robot, OTT Whisper, OTT Glass, OTT Metal, OTT Hollow, Hard Squash, Air Pressure, Dense Broadcast, Final Boss Vocoder, Vintage Eight, Analog Twelve, Studio Sixteen, Speech Twenty, Clear Twenty Four, Modern Thirty Two, Detail Forty, Hi Fi Forty Eight, Precision Fifty Six, Surgical Sixty Four, Dual Anchor Saw, Power Trio Saw, Five Voice Saw, Twelve Voice Saw, Saw Pulse Blend, Balanced Pulse, Bright Pulse Stack, Narrow Pulse Stack, Fifth Up Carrier, Fifth Down Carrier	Init	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Range High VST3 ID 281556343	2000 to 18000	14000	Automatable	Host-automatable control for Range High. Its full range and default are shown in this table; use it with the related feature section above.
Range Low VST3 ID 1810201823	40 to 500	90	Automatable	Host-automatable control for Range Low. Its full range and default are shown in this table; use it with the related feature section above.
Release VST3 ID 1090594823	5 to 2000	50	Automatable	Sets how long the associated detector, envelope, reverb, or resonant process takes to recover or decay.

Parameter	Range / options	Default	Host status	Function
Shift VST3 ID 109407362	-24 to 24	0	Automatable	Moves pitch, frequency structure, or perceived resonant size for the named process.
Sibilance VST3 ID 797472958	0.0 to 100.0	50.0	Automatable	Host-automatable control for Sibilance. Its full range and default are shown in this table; use it with the related feature section above.
Sidechain VST3 ID 302364618	Off, On	Off	Automatable	Host-automatable control for Sidechain. Its full range and default are shown in this table; use it with the related feature section above.
Tone VST3 ID 3565938	-100.0 to 100.0	0.0	Automatable	Host-automatable control for Tone. Its full range and default are shown in this table; use it with the related feature section above.
Type VST3 ID 3575610	Vocoder, Whisper, Morph, Ring Mod, Invert	Vocoder	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Warp VST3 ID 3641992	-100.0 to 100.0	0.0	Automatable	Moves pitch, frequency structure, or perceived resonant size for the named process.
Whiten VST3 ID 1358674277	0.0 to 100.0	75.0	Automatable	Host-automatable control for Whiten. Its full range and default are shown in this table; use it with the related feature section above.
Width VST3 ID 113126854	0 to 200	100	Automatable	Sets stereo spread or the lateral distribution of the processed signal.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.